Eg: Data .

S: 210.76.216.111

D: 19.254.186.113

The network ID of this data is already there in the routing table, so data will be unicasted to e_2 alone.

Day-8: LAN CONFIGURATION

Connection b/w PC & Router → Console cable.

→ Serial console Cable

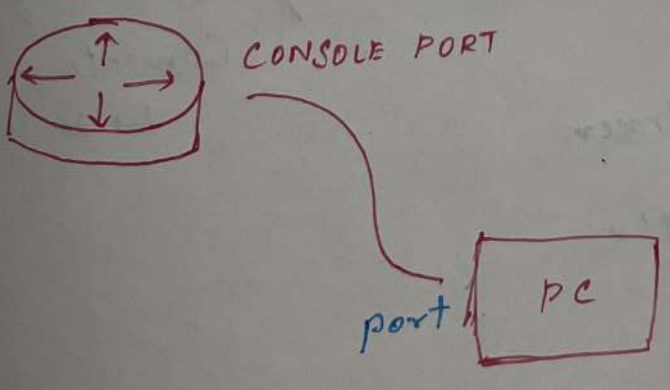
→ USB console Cable.

This cannot be directly connected to PC. USB connector can be used.

CISCO PACKET TRACER:

→ Router - 2620 / 2621 - preferred.

→ If we add extra ethernet module then extra interface is added.



→ If real device is used, download Putty.

After connecting console cable, → device manager

→ COM.

→ If cisco PT is used, PC → Desktop → CMD → OK

→ Would you like to start initial config : no.

Router modes:

1) Router > → user execution mode [Testing]

2) Router # → Privilege mode [view]

3) Router (config)# → Global configuration mode. [Configuration]

$$\text{Router >} \underset{\text{disable}}{\overset{\text{enable}}{\rightleftarrows}} \text{Router\#} \underset{\text{exit}}{\overset{\text{configure terminal}}{\rightleftarrows}} \text{Router (config)\#}$$

? → Gives all the commands.

Show ip interface brief → command to open interface.

UTP cable
↳ network
Cform

Console
↳ to configure

Practicals:

Step-1: Open Cisco packet tracer

2: Select Router - 2621x

3: Select a PC

4: Select console cable.

PC → RS 232 port

Router → console port

5: Go to PC → Desktop → Terminal → no

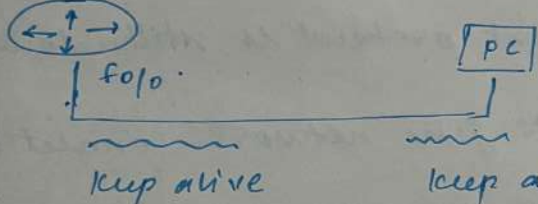
6: Then Router interface opens.

7: Router > Show ip interface brief command

Interface	Status	Protocol
1.	Ad down	down.
	↓	
2.	ON	
3.		

This protocol checks the cable connection

Once the interface is ON, interface will send dummy signals keep alive and the interface from PC will also send keep alive signal



Once the signals are exchanged, protocol is UP.

8: Configuration of Router interface.

→ Go to Global configuration mode

→ Router (config)# interface fastethernet 0/0

↓

This is the name of the interface we are configuring

→ Router (config-if)#

↙

4th mode - Interface mode

9: Give IP address

→ Router(config-if)# ip address 50.0.0.1 255.0.0.0

→ Router(config-if)# no shutdown

↓

to on the interface

10: Go back & check interface again.

→ Router(config-if) # exit

→ Router (config)# exit

→ Router # show ip interface brief.

Interface	ip	Status	Protocol
Fast ethernet 0/0	50.0.0.1	UP	down.

11: IP is configured but protocol is still down because we have to give network connection to PC also, then only it will return the keepalive signals.

12: Give a cross cable connection. Now if we check the interface, protocol is UP.
Use ncpa.cpl command to configure ip in PC in real.

Authentication:

Login local → Username & password

login. → only password.

Steps for including authentication:

1: Goto Router config mode & type

→ Router(config)# username bugs password bugs

→ Router(config)# line console 0

→ " # login local

Here after giving exit & starting again it will ask username & password.

2: Router(config)# line console 0

" password bugs

" login.

Here if we start again only password will be asked directly.

3: Level wise authentication

→ Router(config)# enable secret ccna

For going from Router> to Router# password will be asked now.

This is the authentication process.

login local → Username & PW

login → PW

enable secret xxxx → Level wise authentication

TAKING ROUTER AS REMOTE:

If we are taking router remotely, it is called as telnet

Step-1: Forming network connection b/w PC & Router

2: We should enable the telnet protocol

→ Router (config)# line vty 0 4

↓

5 devices can take the router remotely

→ Router (config-line)# transport input telnet

→ " # login local

" # exit

3: Go to PC → Putty application

→ telnet → Enter Router IP.